

ML Engineer with hands-on experience building end-to-end ML systems across deep learning, statistical modeling, and large language models. Experienced in model benchmarking, feature engineering, and A/B testing, with production-level work on GPU inference platforms and scalable ML pipelines. Adept at applying the right modeling approach — from XGBoost to Transformers — to drive measurable outcomes.

SKILLS

Languages	Python, SQL, C/C++, JavaScript, Bash
ML / AI Libraries	PyTorch, TensorFlow, XGBoost, LangChain, Hugging Face, scikit-learn, NumPy, Pandas, Transformers
Engineering Tools	Docker, Git/GitHub, Databricks, MLflow, NVIDIA Triton, Linux, GCP

EDUCATION

Master of Science in Applied Computing (CGPA 4.0/4), University of Toronto	September 2024 - December 2025
Artificial Intelligence Focus	Toronto, ON, Canada
Honors Bachelor of Science with High Distinction (CGPA 3.94/4), University of Toronto	September 2018 - April 2024
Specialist in Computer Science, Major in Mathematics, Minor in Statistics	Toronto, ON, Canada

TECHNICAL EXPERIENCE

Data Science Intern PyTorch, OpenCV, XGBoost, Databricks, Docker, NVIDIA Triton	May 2025 – December 2025
Magna New Mobility	Toronto, ON, Canada
- For ultra-high-resolution industrial images where full-image inference was impractical, built a patch-based segmentation pipeline that split inputs into windows and stitched predictions, enabling production-scale inference. - To improve detection of small and subtle surface defects, applied targeted data augmentation, model tuning, and optimized stitching strategies, achieving +0.19 IoU, +0.30 Precision, and +0.22 Recall across defect classes. - For leather defect analysis requiring interpretability and flexibility, used a hybrid CV–ML workflow (binary segmentation → defect crop extraction → XGBoost severity classification), producing modular and interpretable predictions. - To support end-to-end deployment and iteration, built Databricks-based ETL pipelines and deployed GPU inference with Docker and NVIDIA Triton, reducing setup time and enabling reproducible training and deployment workflows.	
Software Engineer Intern C++, Python, ROS2 Foxy, OpenCV, Linux, Git	May 2022 - June 2023
Magna Electronics	Brampton, ON, Canada
- For vision-based sensing systems requiring accurate perception, performed intrinsic and extrinsic camera calibration to ensure reliable sensor alignment and measurement accuracy. - To improve robustness of autonomous sensing software, resolved bugs and extended C++/Python modules, strengthening system stability and reliability. - To support customer releases and platform integration, managed version-controlled deployments for customer delivery and implemented ROS2 Foxy C++ features on Linux, while maintaining code quality via testing, reviews, and GitHub workflows.	

PROJECTS

Artwork Restoration with Generative Image Inpainting Python, PyTorch, git	September 2024 - December 2024
- Designed and executed a controlled benchmarking study of 3 generative inpainting models (CO-MOD-GAN, ICT, LaMa) on WikiArt, training on 20k paintings and evaluating on a 5k-image test set with 5 degradation masks. - Quantitatively compared restoration quality using MSE, MS-SSIM, and FID, identifying clear trade-offs: LaMa achieved strongest pixel and structural fidelity (MSE 0.01002, MS-SSIM 0.79205), while Co-Mod-GAN produced the most realistic global distributions (FID 112.155). - Performed qualitative error analysis across multiple art styles to surface practical failure modes, including facial distortion, fine-detail loss, and breakdowns under full occlusion.	
INTR: Interpretable Transformer for Image Classification Python, PyTorch, git	September 2024 - December 2024
- Conducted large-scale evaluation of INTR on 14 fine-grained datasets (8 original + 6 extended), combining accuracy benchmarking with attention heatmap analysis. - Analyzed generalization behavior by comparing INTR against ResNet-50, observing up to +9.2% accuracy gain on Fish and up to -12% degradation on CUB under complex background conditions. - Investigated class-specific cross-attention patterns to assess interpretability, demonstrating INTR’s ability to localize discriminative regions driving fine-grained predictions.	
3D Object Tracking & Motion Forecasting Python, PyTorch, git	December 2021 - April 2022
- Developed LiDAR-based vehicle detection, tracking, and motion forecasting pipelines for autonomous driving perception, under the supervision of Prof. Raquel Urtasun. - Explored robustness improvements in dynamic scenes through hard target mining, Gaussian target representations, and GNN-based modeling, evaluating their impact on tracking stability and forecasting accuracy.	